

Ambient Witness: Repurposing the Language Barrier as a Covert Safety Net in Domestic and Workplace Conflicts

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Abstract

Foreign residents and multicultural families face heightened risks where language barriers, physical surveillance, and power imbalances constrain their ability to seek help. In immediate threats, interacting with a mobile device can be unsafe or impossible, leaving individuals unable to signal distress. This paper examines how language barriers can be repurposed as a covert interaction mechanism, allowing users to embed native-language rescue keywords within ordinary conversation. We implemented an ambient, offline speech recognition device to trigger silent alerts and capture contextual audio. In a formative evaluation with international residents ($N = 8$), participants described hands-free, linguistically camouflaged interaction as superior to conventional safety apps in high-stakes conflicts. Participants also identified design directions, including multimodal triggers and smaller form factors. Our findings suggest that treating language barriers as layered communication channels opens new directions for designing discreet, user-centered safety systems.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing systems and devices; *User studies*; • **Security and privacy** → Domain-specific security and privacy architectures.

Keywords

Ambient Computing, Personal Safety, Language Barriers, Covert Interaction, Domestic Violence Prevention

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1 Introduction

Finding help is a complex, non-linear process involving recognizing a problem, making the decision to seek assistance, and selecting a support provider [10]. For immigrants and multicultural families in foreign countries, this procedure is further complicated by language barriers, social isolation, and physical limitations. In high-stakes situations such as intimate partner violence, the primary barrier is often not the absence of technology, but the lack of safe, unmonitored access to it [7, 17]. This creates a dangerous vulnerability where the primary tool of safety is physically out of reach at the moment of greatest need.

Current digital safety interventions often assume a level of privacy and autonomy that victims do not have; in practice, high-risk encounters involve physical constraints and potential threat that make such interactions unsafe [7, 17]. Previous work on mobile safety applications highlights a dependence on GPS tracking and manual SOS triggers, which require sustained attention and physical activation [15]. While “quick exit” buttons provide a mechanism for users to hide their digital activity in real-time, the physical world lacks an equivalent that can immediately shield one from the actions of an aggressor [18].

In this paper, we investigate how language barriers can function as a defense mechanism rather than a limitation. We explore how individuals can embed native-language rescue keywords into ordinary conversation to trigger help without alerting the aggressor. We assessed the applicability of this idea using ambient, offline speech recognition to detect keywords and trigger silent alerts while capturing contextual audio.

We report findings from a formative evaluation with international residents ($N = 8$) in South Korea, focusing on how participants envisioned using hands-free interaction, linguistic camouflage, and perceived safety in high-stakes situations. Our study demonstrates how leveraging one’s cultural identity and language can transform a barrier into a shield, providing a lifeline when the hands are tied and the cameras are blind.

2 Related Work

2.1 Sociocultural Vulnerability and Coercive Control

Foreign residents in rapidly diversifying societies may encounter structural and social barriers that complicate integration [9]. In South Korea, for instance, the increase in multicultural households and foreign workers has been accompanied by persistent anti-foreign sentiment [9, 13]. This systemic exclusion is quantified by recent survey data, which reveals that 43.7% of foreign residents report experiencing discrimination, most frequently caused by their country of origin or limited Korean language proficiency [12]. This vulnerability can be weaponized through “Technology-Enabled Coercive Control” (TECC), where digital technologies are used to monitor, isolate, and intimidate victims [6]. For foreigners who may face additional linguistic or legal obstacles, standard tools may be insufficient, particularly where victims lack the agency to act on surveillance alerts [1]. As such, smart technologies must be designed to prevent misuse of monitoring and coercive control, particularly in contexts where users face structural vulnerabilities. [11].

2.2 From Mobile Limitations to Stealth Interventions

Current safety applications can be difficult to use safely in high-risk environments, as they depend on tactile interaction and manual SOS activation [15]. These designs assume safe access to the device, which can be compromised during periods of surveillance or coercive control [3]. Because visible help-seeking actions can increase risk, there is a need for more discreet and less detectable safety interventions [4]. Although digital “quick exit” mechanisms effectively hide sensitive web content [18], these stealth principles are rarely applied to physical environments where device access is constrained. Our work extends these principles beyond the screen by using native-language voice activation as a shield, providing a safety net that operates independently of the control of the abuser.

2.3 Limitations of Reactive Safety Tools

Wearable hardware such as inVisaWear offers a dedicated SOS trigger [8], but still requires a deliberate physical motion that can raise suspicion under observation. Emergency sharing platforms such as Noonlight rely on the user’s ability to navigate a mobile interface [14], which is a significant barrier when a device is physically out of reach. While “vault” applications provide a secure environment for sensitive data [16], they primarily function as tools to preserve evidence rather than offer immediate protection. These options highlight a distinct gap during active confrontations, where manual signaling remains a risk. There is a critical need for safety mechanisms that operate effectively without requiring tactile engagement or visible device interaction.

3 Formative Study

To understand how foreign residents currently manage safety concerns, we conducted semi-structured interviews with three long-term migrant women living in South Korea. The participants were

recruited through word of mouth; one of them identified as Korean-Chinese (Chaoxianzu) holding an F-4 visa, while the remaining participants were Chinese nationals who have resided in Korea for 15 to 20 years. Each session lasted an hour and questions were framed to understand their life experiences in a foreign country.

Paralyzed by Surveillance. Participants revealed that in high-stakes environments, such as hostile workplaces or domestic conflicts, the primary constraint is not a lack of network connectivity, but the constant presence and monitoring of an aggressor. A participant who works in a food processing plant described her workplace as an environment in which mobile phone use is strictly prohibited and actively monitored. When faced with verbal harassment from colleagues, she lacked any means to document incidents or request assistance without violating the rules of the workplace. This illustrates a limitation of current safety tools, which assume uninterrupted access to a mobile device.

The Risk of Escalation. Participants also described situations in which an overt interaction with a phone could escalate conflict. One interviewee noted that in both workplace and family settings, the use of devices can provoke the other party, increasing the risk of retaliation. In such contexts, the simple act of attempting to seek help can exacerbate harm. These accounts suggest that an effective safety mechanism must operate without requiring explicit device manipulation or visible signaling.

Beyond the Home. Our research initially focused exclusively on the intervention of domestic violence in multicultural families. However, the formative interviews revealed that the primary issue is the inability to safely manipulate a communication device under conditions of surveillance, restriction, or increasing tension. This constraint applies throughout domestic and workplace settings, affecting individuals whose movements, attention, or behavior are closely monitored.

The interviews have expanded the scope beyond the home and toward the design of a ubiquitous hands-free safety mechanism. Rather than framing the intervention as a conventional panic button, we explore the role of ambient systems that can function as discreet witnesses, enabling requests for assistance without overt interaction. This re-framing directly informs the interaction mechanism examined in this paper.

4 System Design

4.1 Design Rationale

Existing safety tools often fail because they require visible physical interaction, which can provoke and escalate the conflict. To enable covert interaction, we selected ambient speech as the primary modality. This allows users to request assistance without explicit device manipulation or visual attention. By embedding rescue keywords in a language not understood by the others present, users can trigger the system while maintaining the appearance of ordinary conversation, prioritizing discretion in environments where explicit signaling is unsafe.

Table 1: Participant demographics and selected rescue keywords

ID	Gender	Age	Nationality	Affiliation	Native Language	Keyword	Meaning / Context
P1	M	20–25	India	Student	Hindi	"Bachao"	Help
P2	F	20–25	Vietnam	Student	Vietnamese	"Mẹ"	Mother
P3	F	25–30	Vietnam	Researcher	Vietnamese	"Mẹ"	Mother
P4	F	20–25	Indonesia	Student	Indonesian	"Oven"	Kitchen
P5	M	30–35	Vietnam	Researcher	Vietnamese	Generic	–
P6	F	25–30	Vietnam	Student	Vietnamese	Generic	–
P7	M	18–20	Indonesia	Researcher	Indonesian	"Car"	Car
P8	M	18–20	India	Student	Hindi	"Bachao"	Help

4.2 System Overview

The system is a stationary emergency support prototype designed for environments where the use of mobile phones may be restricted or monitored. Intended for placement in high-risk domestic or shared spaces (e.g. the kitchen, living room, or office), it operates via an always-on background process with offline speech recognition to detect user-defined rescue keywords. When activated, the system records a short segment of contextual audio and sends a covert alert to a trusted contact. All audio processing occurs locally, and no conversational data leaves the device unless a keyword is detected.

4.3 Implementation

Configuration: During a safe period, the user registers a rescue keyword on a frontend interface (React). The keyword is typically chosen from the user’s native language or as a phrase unlikely to occur amidst a conversation with locals. **Passive Monitoring:** The system continuously processes ambient audio using offline speech recognition (VOSK) running locally on a Raspberry Pi. The lightweight language models (approximately 50MB) are suitable for real-time inference and support more than 20 languages, allowing the system to be tailored to the native language of the target user [2]. The audio is examined only for the presence of the rescue keyword and is discarded immediately if no match is found. **Activation:** In a high-risk situation, the user speaks the rescue keyword naturally within the ongoing conversation. No physical interaction or visual feedback is required, with the detection activating in approximately 2 seconds. **Response:** Upon keyword detection, the system records a fixed-duration audio segment and sends an alert via email to a pre-designated trusted contact. This provides the recipient with immediate contextual information while minimizing unnecessary data transmission.

5 Evaluation

To validate the effectiveness of the proposed “hands-free” interaction model, we conducted a controlled, scenario-based study. The goal was not to simulate experiences of abuse but to assess the reliability of keyword-based activation under natural speech conditions and participants’ perceived sense of security compared to conventional mobile safety tools.

5.1 Participants

We recruited participants ($N = 8$) from the international community at our institution (located in South Korea) using snowball sampling.

All were non-native Korean speakers from Vietnam, Indonesia, and India, several of whom reported prior experiences of cultural and social isolation. The study was conducted in a controlled laboratory setting where informed consent was obtained and participants were briefed on the system’s always-on listening behavior, with the assurance that no audio data were retained after the session. Participants received a modest honorarium (approximately USD 10 in local currency) for their participation (Table 1).

5.2 Procedure

Each session consisted of the following three phases:

Keyword Setup. Participants were first introduced to the system and asked to select a rescue keyword that felt natural to use in everyday conversation, often drawn from their native language or from a contextually plausible term (e.g., “Mẹ,” “Oven,” “Bachao,” “Car”). Participants also set the recording duration (30–60 seconds) to determine how long the audio would be captured after activation.

Activation Testing. The participants then tested the system by embedding their chosen keyword in natural sentences while conversing with the researchers in English. They were asked to vary their pace and tone, including casual speech and more urgent delivery, to assess robustness under different vocal conditions. Upon activation, participants continued the conversation for the remainder of the recording period and then verified receipt of the audio file via email.

User Feedback. Following the interaction, we conducted an interview to understand participants’ perceptions of the interaction. Participants completed a short survey using a 5-point Likert scale to assess ease of configuration, intuitiveness, perceived safety relative to smartphone-based tools, and confidence that activation would remain covert. Open-ended questions were asked to raise concerns, suggestions, and applicability beyond our current system.

6 Results

Participants evaluated the system using a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). Overall, participants rated voice-based activation as significantly safer than smartphone interaction ($M = 4.8$, $SD = 0.45$), preferring hands-free mechanisms for high-risk contexts. Confidence that the native-language trigger would remain unnoticed was also high ($M = 4.6$, $SD = 0.55$), validating the use of language barriers as effective camouflage. While the always-on model enhanced perceived security, lower ratings for control and flexibility suggest a need for more granular response options, such as tiered alert levels or cancellation features.

6.1 Qualitative Insights

Reducing Physical Risk. Participants consistently identified the absence of physical interaction as the system’s primary advantage over mobile safety applications. P1 noted that in conflict situations, “you may not find your phone around, or making an action [to reach for it] could be more suspicious.” P6 described the interaction as “verbal, easy,” while P4 emphasized that voice activation felt “safer and more comforting” because it avoided visible actions that could escalate conflict.

Language Barrier as a Protective Layer. Participants leveraged linguistic difference to maintain plausible conversation while signaling for help. Keywords such as “Bachao” (Hindi for help) or context-specific terms like “Oven” were perceived as effective camouflage. P8 explained that “the chance of him knowing my language would be really low... so the word itself wouldn’t cause any more stress.” P6 similarly reported feeling “no anxiety” when activating the system during the mock interaction.

Towards Miniaturization. Participants identified several areas for refinement. P6 suggested supporting multiple keywords tied to different levels of urgency, along with a cancellation or confirmation mechanism to prevent false alarms, a concern echoed by P2. Others proposed redundancy through multiple trigger words (P7), multi-modal extensions such as a “fake call” or location sharing (P4, P3), and more discreet or wearable form factors to extend protection beyond fixed locations (P5, P7).

7 Discussion

Our study validates that removing the need to interact with a physical device reduces the burden on victims during a crisis. Participants identified the hands-free nature of the system as a critical advantage, noting that the location and unlocking of a smartphone is a dangerous liability during conflict. Voice activation provided emotional relief by bypassing the need to conceal actions from an aggressor. This suggests non-tactile safety tools can alleviate the anxiety of being monitored and empower users to act rather than remain passive.

The interviews also confirmed the effectiveness of the use of language barriers as a security feature. Using native-language keywords allowed participants to embed rescue requests into natural conversation without raising suspicion. This transforms linguistic isolation from a vulnerability into a mechanism for covert signaling. By leveraging cultural identity as a shield, the system offers a layer of security that standard panic buttons do not provide.

Our system addresses the critical need for discreet intervention and proactive safety [15], specifically in environments where survivors are sequestered with abusers and may lack private access to their devices. We extend “quick exit” principles from web browsers into the physical domain [18], creating a discreet witness that requires no physical contact. Unlike existing apps that require visual focus and manual dexterity, our system operates peripherally. This allows the user to maintain social and physical situational awareness while triggering a rescue protocol. By employing a local network and processing pipeline, our system mitigates the data and surveillance risks inherent in cloud-connected IoT devices. [1].

8 Limitations and Future Work

A primary limitation of the current prototype lies in its hardware form factor. The Raspberry Pi-based implementation is bulky and restricts deployment to fixed domestic environments, creating a protection gap once the user leaves the home. This constraint surfaced when considering use beyond private spaces, particularly in workplaces where harassment occurs under continuous observation.

While our formal evaluation focused on international students, supplementary interviews with Vietnamese service workers offered an important perspective on this limitation. Although these interviews did not follow the complete study protocol, the participants described workplace harassment experiences and emphasized that the system would be most viable if it were adapted in a discreet and portable form. Suggestions included wearable formats, such as a pendant, that could be worn during restaurant shifts without drawing attention. This feedback indicates that mobility and concealment are necessary for real-world adoption, especially in customer-facing service roles.

Miniaturization of hardware is the key to transitioning the system from a stationary appliance to a portable or even wearable device, extending protection to workplaces and public transportation. Participants also requested greater control over system behavior, including cancellation mechanisms to prevent false positives and features such as a fake incoming call that could provide a socially plausible exit from escalating situations. By integrating these enhancements, the system can move beyond a proof-of-concept toward a practical safety tool that provides the objective data necessary for facilitators to verify accounts of abuse [4]. This approach directly supports workplace and domestic abuse interventions by replacing generic training with real evidence-based insights [5].

9 Conclusion

Foreign residents and multicultural families face safety risks where surveillance, power imbalance, and language barriers limit access to help. Existing mobile safety tools often fail in these situations by relying on visible, manual interaction that can escalate harm. This work explored an alternative model that uses linguistic differences as an asset rather than a constraint. We presented an ambient, offline speech recognition prototype that allows users to embed native-language rescue keywords within ordinary conversation. An evaluation with international residents indicates that hands-free activation and linguistic camouflage were perceived as safer and more plausible than smartphone-based approaches. By reframing language barriers as a covert interaction channel, this work suggests design opportunities for safety systems that must operate under surveillance and restricted device access.

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